

Multiple Choice (10 marks)

- Which of the following lists the hydrides of group-14 elements in order of thermal stability, from lowest to highest?
 - $\text{PbH}_4 < \text{SnH}_4 < \text{GeH}_4 < \text{SiH}_4 < \text{CH}_4$
 - $\text{PbH}_4 < \text{SnH}_4 < \text{CH}_4 < \text{GeH}_4 < \text{SiH}_4$
 - $\text{CH}_4 < \text{SiH}_4 < \text{GeH}_4 < \text{SnH}_4 < \text{PbH}_4$
 - $\text{CH}_4 < \text{PbH}_4 < \text{GeH}_4 < \text{SnH}_4 < \text{SiH}_4$
- What is the limiting high-temperature molar heat capacity at constant volume (C_V) of a gas-phase diatomic molecule?
 - 1.5 R
 - 2 R
 - 2.5 R
 - 3.5 R
- The T_2 of an NMR line is 15 ms. Ignoring other sources of linebroadening, calculate its linewidth.
 - 0.0471 Hz
 - 21.2 Hz
 - 42.4 Hz
 - 66.7 Hz
- The normal modes of a carbon dioxide molecule that are infrared-active include which of the following? I. Bending II. Symmetric stretching III. Asymmetric stretching
 - I only
 - II only
 - III only
 - I and III only
- Considering 0.1 M aqueous solutions of each of the following, which solution has the lowest pH?
 - Na_2CO_3
 - Na_3PO_4
 - Na_2S
 - NaCl

True / False (10 marks)

Answer True or False

6. True or False: Protons in a molecule with a rotational correlation time of 1 ns have the fastest spin-lattice relaxation at a magnetic field of 3.74 T.
7. True or False: The nuclide ^{19}F has an NMR frequency of 115.5 MHz in a 20.0 T magnetic field.
8. True or False: AgCl has a lower melting point than HCl because it is an ionic compound with strong ionic bonds.
9. True or False: Most organic compounds do not contain silicon atoms, making tetramethylsilane an ideal reference for ^1H NMR spectroscopy.
10. True or False: The molecular geometry of thionyl chloride, SOCl_2 , is tetrahedral because it has four bonding pairs of electrons.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors influencing the strength of a ^{13}C 90° pulse of duration 1 s, and calculate its value, provided the pulse is applied to a sample of ^{13}C at 125 MHz.
11. Discuss the concept of dissociation energy in the context of the hydrogen-bromine bond, specifically analyzing the enthalpy change associated with the reaction $\text{HBr(g)} \rightarrow \text{H(g)} + \text{Br(g)}$.

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